

Ema Maria Monteiro Martins

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Master's student in Informatics and Computing Engineering with experience across **backend**, **frontend**, and **mobile** development. Skilled in **C#**, **Python**, and **JavaScript**, with hands-on experience building **full-stack applications**, **real-time systems**, and **AI-integrated software**.

Education

Universidade do Porto	Sep 2024 - Jul 2026
Master's Degree in Informatics and Computing Engineering	Porto, Portugal
Universidade do Minho	Sep 2022 - Jul 2024
Bachelor's Degree in Informatics Engineering (completed concurrently with Biomedical Engineering)	Braga, Portugal
Universidade do Minho	Sep 2020 - Jul 2024
Bachelor's Degree in Biomedical Engineering (completed concurrently with Informatics Engineering)	Braga, Portugal

Experience

Mindera	Feb 2026 - Jun 2026
Master Thesis Intern	Porto, Portugal
- Developed an internal project engineering an AI agent to act as a believable second player in two-player video games, prioritizing human-like behavior over flawless performance. - Integrated computer vision and AI models (NVIDIA NitroGen, YOLO, Grounding DINO) to process in-game environments in real-time.	

Projects

Finalyze React Native, Spring Boot, PostgreSQL	Repo Link
- Collaborated on the strategic and technical development of Finalyze, a fintech mobile app designed to help users track expenses, set savings goals and build financial habits through gamification and data-driven benchmarking. - Co-authored key strategic documents (Business Model Canvas (BMC), Value Proposition Canvas (VPC), Product Vision Board, Project Charter and Strategic Plan), following agile project management practices.	
Car Mechanic Interface Vue.js, JSON Server	Repo Link
- Designed and developed an auto repair shop interface tailored for mechanics to streamline daily service workflows. - Implemented core functionality to execute, pause, and track repair services, including logging detailed explanations for task blockers. - Integrated user management features enabling mechanics to access and update personal profile information directly within the system.	
Receitas Kotlin	Repo Link
- Developed a mobile app ("Receitas") to digitize personal recipe management and replace paper-based tracking. - Implemented full CRUD operations allowing users to efficiently add, view, update, and delete custom recipes. - Engineered a JSON export feature to support data portability, backups, and seamless integration with other applications.	
Clothing Recommendation System Python Flask	Repo Link
- Developed a Clothing Recommendation System using a custom KD-Tree algorithm and Euclidean weighting, matching suggestion quality of major e-commerce platforms such as Primark and Shein. - Built a full-stack Flask web application, optimizing high-dimensional one-hot encoded data for fast, transparent product retrieval.	

Skills

Technical Skills	Languages
Backend: C, C#, C++, Python, Java, Haskell, .NET	Portuguese: Native
Frontend: JavaScript, React, Vue.js, HTML/CSS, Pug	English: Fluent
Mobile: Kotlin, React Native, Flutter	Spanish: Basic
Databases: MongoDB, MySQL, Redis, PostgreSQL, Neo4j	
Practices: Git, Agile/Scrum	
Game Dev: Godot, Unity	
AI & Machine Learning: Unity ML-Agents, NitroGen, YOLO, Grounding DINO	

Leadership & Volunteering

Volunteer – Extracurricular Classes	Sep 2022 – Mar 2023
Volunteering: Taught extracurricular classes to children aged 6–9, promoting environmental awareness.	Braga, Portugal
Vice-president of Literatuna – Universidade do Minho	Sep 2023 – Jul 2024
Student Organization Leadership: Coordinated activities and performances of a university musical group.	Braga, Portugal
Erasmus+ Project – ANIMA 2022	May 2022
International Sociocultural Event: International meeting of sociocultural animation.	Portugal / Europe